

Summary

1. Introduction

Graphics and visual computing focus on the creation, processing, and visualization of images and scenes using computers. This field bridges art and technology, allowing us to represent information visually and build immersive digital experiences. In today's world, its importance is seen everywhere—from video games, movies, and design software to medical imaging, simulations, and virtual reality—making it one of the most influential areas of computing.

2. Historical Progression

The progression of graphics and visual computing has been shaped by several milestones. It began with early **vector graphics and raster displays**, supported by the development of **frame buffers** that enabled image storage and manipulation. As technology advanced, **full-color games** emerged, highlighting the shift from simple visuals to more vibrant displays. Two paradigms—**raster and vector**—dominated early systems. Innovations such as **smooth scrolling** and **sprite scaling** (resizing images) enhanced interactivity, with **sprites** becoming central in 2D games. The era of **parallax scrolling** gave depth to flat visuals, while **cartoon classics** and **rotoscoped animation** introduced more natural and engaging movements. The **multimedia advent** opened the door for richer digital content. Entering the **polygon realm**, early 3D graphics faced hardware limitations, often relying on **flat shading** for simplicity. The introduction of **real-time 3D graphics** through **hardware acceleration and GPUs** was a turning point, further boosted by the iconic **Voodoo graphics card**, which offloaded rendering from CPUs. New styles and effects followed, including **cell shading**, **motion blur**, **bloom lighting**, and immersive **first-person perspectives**, all of which pushed realism and artistic expression forward.

3. Key Concepts and Learnings

From the video, I learned that graphics evolution is not just about visual beauty but about solving technical challenges to balance performance and realism. Concepts like scaling, shading, and hardware acceleration show how each leap opened creative possibilities for developers and artists. The role of GPUs stands out as a critical enabler of modern 3D rendering, shaping industries beyond gaming, such as film, simulation, and AI-powered imaging.

4. Personal Reflection

The most fascinating part for me was seeing how simple tricks like parallax scrolling and rotoscoping once felt revolutionary and laid the groundwork for today's realistic visuals. Looking ahead, I believe the field will continue to merge with artificial intelligence, enabling **neural rendering**, **real-time photorealism**, and **immersive AR/VR experiences**, making graphics more accessible and lifelike than ever before.

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